Hand Derivation 2: Optimal Voluntary Buffer

1. Banks Objective

$$\min_{\mu_K \in [0, \bar{\mu}_K]} J_K(\mu_K)$$

where

$$J_K(\mu_K) = r\mu_K + E_A\left[i\phi_K \frac{D_K}{\partial K} - i l_K \frac{\chi_K}{\partial K} + K(1-\phi)(1-\psi_K)\frac{D_K}{\partial K} \right]$$

The bank takes the state-contingent market schedules and prices as given

Define

$$a_K(A) = i\psi_K(A) + K[1-\phi(A)][1-\psi_K(A)]$$
$$b_K(A) = i l_K(A)$$

Then

$$J_K(\mu_K) = r\mu_K + E_A[a_K L_K - b_K M_K]$$

2. First derivative

$$J_K'(\mu_K) = r + E_A[a_K L_{K,\mu} - b_K M_{K,\mu}]$$
$$= r + E_A[-a_K \pi_K^- - b_K \pi_K^+]$$
$$= r - E_A[a_K \pi_K^- + b_K \pi_K^+]$$

3. Inner FOC

If

$$0 < \mu_K^* < \bar{\mu}_K$$

then

$$J_K'(\mu_K) = 0$$

Therefore

$$r - E_A[a_K \pi_K^- + b_K \pi_K^+] = 0$$
$$r = E_A[a_K \pi_K^- + b_K \pi_K^+]$$

Replacing

$$r = E_A[(i\psi_K + K(1-\phi)(1-\psi_K))\pi_K^- + i l_K \pi_K^+]$$

Marginal carrying cost = avoided default cost + additional interbank lending income

4. Second derivative

(with conditional density $f_{\omega_K|A}$)

$$\frac{\partial \pi_K^-}{\partial \mu_K} = -f_{\omega_K|A}(-\mu_K)$$
$$\frac{\partial \pi_K^+}{\partial \mu_K} = f_{\omega_K|A}(-\mu_K)$$

then

$$J''(\mu_K) = -E_A\left[a_K \frac{\partial \pi_K^-}{\partial \mu_K} + b_K \frac{\partial \pi_K^+}{\partial \mu_K}\right]$$

$$= E_A[a_K f_{\omega_K|A}(-\mu_K) - b_K f_{\omega_K|A}(-\mu_K)]$$

$$J''(\mu_K) = E_A[(a_K - b_K) f_{\omega_K|A}(-\mu_K)]$$

5. Convexity and corners

$$a_K(A) \geq b_K(A) \quad \forall A$$
$$\Rightarrow J''(\mu_K) \geq 0$$

Thus

State-by-state $a_K \geq b_K$ is sufficient for convexity

But if $\phi = 1$

$$a_K - b_K = i(\psi_K - l_K)$$

If $i > 0$ and $\psi_K < l_K$ in a positive density state, that state contributes negative curvature

Therefore, the FOC is not automatically sufficient

Under global convexity

$$J'_K(0) \geq 0 \rightarrow 0 \text{ minimizes } J_K$$

$$J'_K(\bar{u}_K) \leq 0 \rightarrow \bar{u}_K \text{ minimizes } J_K$$

Compare all feasible stationary points, relevant kinks and boundary candidates. Uniqueness requires additional conditions.